

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455982.78774 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.012
Transit depth (Rp/Rs)^2	1.97 +/- 0.34 [%]
Orbital Inclination (inc)	83.17 +/- 1.48
Airmass coefficient 1 (a1)	1.1 +/- 0.01
Airmass coefficient 2 (a2)	-0.066 +/- 0.018
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	None
Optimal Aperture	4.28
Optimal Annulus	8.91
Transit Duration (day)	0.0523 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.012 vs 0.1594 (deviation 1.62σ)
Duration fit vs published	0.0523 d vs 0.0512 d (deviation 0.26σ)
Mid-time O–C	0.7 min
Depth SNR	11.7
Residual scatter	0.94 %

Light curve

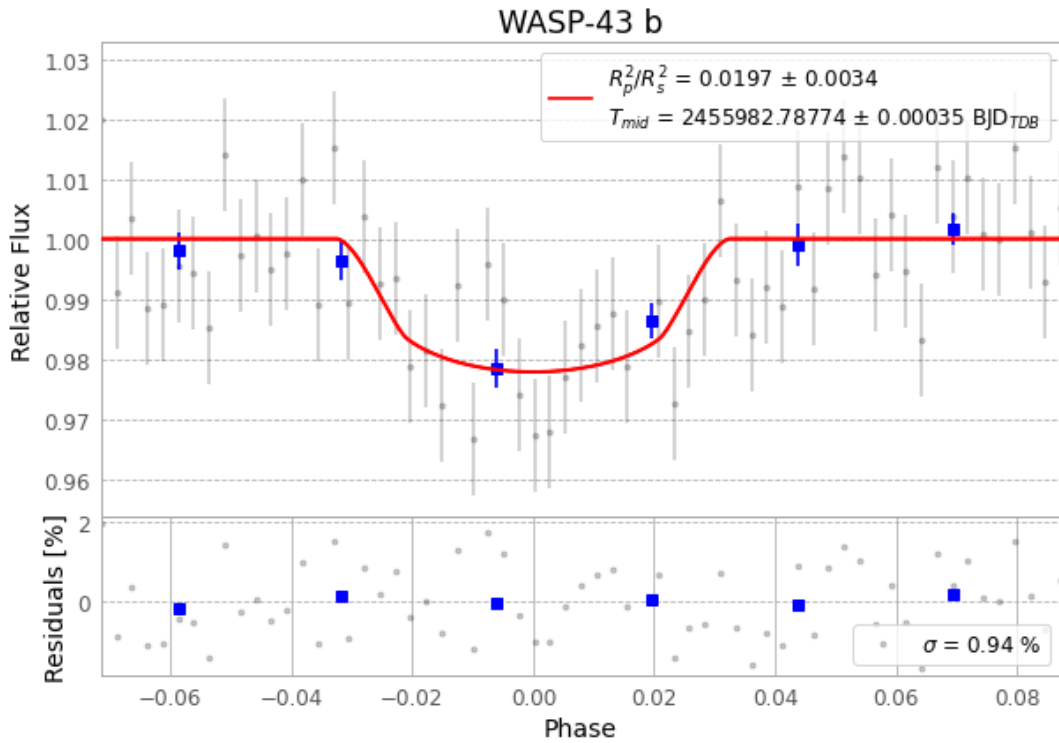


Figure 1 — FinalLightCurve_WASP-43 b_2012-02-24.png (SHA-256 67848e8d4de4c3c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 406143103d20be2e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), WASP-43120225052112.FITS → WASP-43120225082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-120225.log (ad1dfbe6090b...), FinalLightCurve_WASP-43 b_2012-02-24.png (67848e8d4de4...), FinalLightCurve_WASP-43 b_2012-02-24.csv (560feff2e19d...)

Compute resources

Wall time 292.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)